Georgia Tech

MENU

# Master of Science in Analytics

**an Interdisciplinary Hybrid Data Science and Analytics degree**

## Courses

Home › Curriculum

Thanks to Georgia Tech's strengths in each of the key areas of analytics and data science, there are more than 80 courses that MS Analytics students can take to fulfill required and elective slots in their curriculum. Students are encouraged to choose electives to develop specific expertise within an area of analytics/data science where they have career interests.

Courses available to the students either as core requirements or elective options include topics such as machine learning, forecasting, regression analysis, data mining, statistical learning, natural language, computational statistics, simulation, digital marketing [illegible]ization, visualization, databases, web ar[illegible] performance computing, graph [illegible] analytics, revenue management [illegible] analysis, decision support, priva[illegible] below for the full list).

We provide academic and caree[illegible] to help you select the right set o[illegible] interests and career goals.



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The list below is a catalog of electives that are generally offered. Semester availability is subject to change.

# MSA ELECTIVE COURSES

Search

## COMPUTING ELECTIVES

- ▶ CS 3510 - DESIGN AND ANALYSIS OF ALGORITHMS
- ▶ CS 4400 - EMERGING DATABASE TECHNOLOGIES APPLICATIONS
- ▶ CS 4460 - INTRO INFO VISU
- ▶ CS 6035 - INTRO TO INFOR
- ▶ CS 6220 - BIG DATA SYSTE
- ▶ CS 6400 - DATABASE SYSTEMS CONCEPTS AND DESIGN
- ▶ CS 6471 - COMPUTATIONAL SOCIAL SCIENCE



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- CS 6476 - INTRO TO COMPUTER VISION
- CS 6515 - INTRO TO GRADUATE ALGORITHMS
- CS 6550 - DESIGN AND ANALYSIS OF ALGORITHMS
- CS 6601 - ARTIFICIAL INTELLIGENCE
- CS 6730 - DATA VISUALIZATION PRINCIPLES & APPLICATIONS
- CS 6750 - HUMAN-COMPUTER INTERACTION
- CS 7545 - MACHINE LEARNING THEORY
- CS 7632 - GAME AI
- CS 7637 - KNOWLEDGE-BASED AI
- CS 7643 - DEEP LEARNING
- CS 7650 - NATURAL LANGU
- CS 7750 - MATH FND OF M
- CS 8803 - GRAPH NEURAL
- CS 8803 - DEEP LEARNING
- CS 8803 - DATA MANAGEM
- CS 8803 - ADVANCED DB SYSTEM IMPLEMENTATION

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▶ CSE 6010 - COMPUTATIONAL PROBLEM SOLVING FOR SCIENTISTS AND ENGINEERS

▶ CSE 6040 - COMPUTING FOR DATA ANALYSIS: METHODS AND TOOLS

▶ CSE 6140 - COMPUTATIONAL SCIENCE AND ENGINEERING ALGORITHMS

▶ CSE 6220 - HIGH PERFORMANCE COMPUTING

▶ CSE 6230 - HIGH PERFORMANCE PARALLEL COMPUTING: TOOLS AND APPLICATIONS

▶ CSE 6240 - WEB SEARCH & TEXT MINING

▶ CSE 6242 - DATA & VISUAL ANALYTICS

▶ CSE 6643 - NUMERICAL LINEAR ALGEBRA

▶ CSE 6644 - ITERATIVE METHODS FOR SYSTEMS OF [illegible]ATIONS

▶ CSE 6730 - MODELING AND
IMPLEMENTATION

▶ CSE 7750 - MATHEMATICA
LEARNING

▶ CSE 8803 - DATA SCIENCE

▶ CSE 8803 - DEEP LEARNING FOR TEXT DATA ANALYSIS

▶ CSE 8803 - INTRO TO URBAN COMPUTING

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- CSE 8803 - DATA SCIENCE FOR EPIDEMIOLOGY
- CSE 8803 - INTRO TO UNCERTAINTY QUANTIFICATION
- CSE 8803 - MACHINE LEARNING FOR COMPUTATIONAL BIOLOGY
- CSE 8803 - HUMAN IN THE LOOP DATA ANALYTICS
- CSE 8803 - IMAGING WITH DATA-DRIVEN MODELS
- CSE 8803 - ADVANCED NLP
- CSE 8803 - SYSTEMS FOR MACHINE LEARNING

# BUSINESS ELECTIVES

- CS 6725 - INFORMATION SECURITY STRATEGIES AND POLICIES
- MGT 6046 - APPLICATIONS OF DATA ANALYTICS ACCOUNTING
- MGT 6074 - FINTECH & CR
- MGT 6080 - INVESTMENTS
- MGT 6086 - ENTREPRENEU
- MGT 6214 - BUSINESS ANALYTICS PRACTICUM
- MGT 6310 - MARKETING RESEARCH

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- **MGT 6311 - DIGITAL MARKETING**
- **MGT 6314 - UNDERSTANDING MARKETS WITH DATA SCIENCE**
- **MGT 6400 - PRICING ANALYTICS AND REVENUE MANAGEMENT**
- **MGT 6401 - SUPPLY CHAIN MODELING**
- **MGT 6450 - PROJECT MANAGEMENT**
- **MGT 6655 - BUSINESS DATA PREPARATION & VISUALIZATION**
- **MGT 6725 - INFORMATION SECURITY STRATEGIES AND POLICIES**
- **MGT 6726 - PRIVACY, TECHNOLOGY, POLICY, AND LAW**
- **MGT 8803 - FINANCIAL MARKET INSTRUMENTS AND RISK MANAGEMENT**

## STATISTICAL EL

- **CS 7641 - MACHINE LEARN**
- **CSE 6740 - COMPUTATION MINING, AND COMPUTATION**
- **ISYE 6402 - TIME SERIES A**
- **ISYE 6404 - NONPARAMETRIC DATA ANALYSIS**

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▶ ISYE 6405 - STATISTICAL METHODS FOR MANUFACTURING DESIGN AND IMPROVEMENT

▶ ISYE 6412 - THEORETICAL STATISTICS

▶ ISYE 6413 - DESIGN AND ANALYSIS OF EXPERIMENTS

▶ ISYE 6414 - STATISTICAL MODELING AND REGRESSION ANALYSIS

▶ ISYE 6416 - COMPUTATIONAL STATISTICS

▶ ISYE 6421 - BIOSTATISTICS

▶ ISYE 6740 - COMPUTATIONAL DATA ANALYSIS: LEARNING, MINING, AND COMPUTATION

▶ ISYE 6783 - STATISTICAL TECHNIQUES OF FINANCIAL DATA ANALYSIS

▶ ISYE 6805 - RELIABILITY ENGINEERING

▶ ISYE 6810 - SYSTEM MONI

▶ ISYE 7400 - ADVANCED DES

▶ ISYE 7401 - ADVANCED STA

▶ ISYE 7405 - MULTIVARIATE

▶ ISYE 7406 - DATA MINING AND STATISTICAL LEARNING

▶ ISYE 8803 - STATISTICAL MACHINE LEARNING

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▶ **ISYE 8803 - HIGH DIMENSIONAL ANALYTICS**

# OPERATIONS RESEARCH ELECTIVES

▶ **CS 7751 - PROBABILISTIC GRAPHICAL MODELS IN MACHINE LEARNING**

▶ **ISYE 6230 - ECONOMIC DECISION ANALYSIS**

▶ **ISYE 6307 - SCHEDULING THEORY**

▶ **ISYE 6333 - OPERATIONS RESEARCH FO SUPPLY CHAIN ENGINEERING I**

▶ **ISYE 6334 - OPERATIONS RESEARCH FO SUPPLY CHAIN ENGINEERING II**

▶ **ISYE 6644 - SIMULATION**

▶ **ISYE 6650 - PROBABILISTI APPLICATIONS.**

▶ **ISYE 6663 - NONLINEAR OF**

▶ **ISYE 6669 - DETERMINISTI**

▶ **ISYE 6679 - COMPUTATION**

▶ **ISYE 6832 - SIMULATION THEORY AND METHODS**

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- ISYE 7683 - ADVANCED NONLINEAR PROGRAMMING
- ISYE 7686 - ADVANCED COMBINATORIAL OPTIMIZATION
- ISYE 7687 - ADVANCED INTEGER PROGRAMMING
- ISYE 7751 - PROBABILISTIC GRAPHICAL MODELS IN MACHINE LEARNING
- ISYE 8803 - MODELING AND ANALYSIS
- ISYE 8813 - FIRST-ORDER METHODS OF OPT
- ISYE 8813 - CONTINUOUS OPTIMIZATION
- ISYE 8813 - MODERN CONVEX OPTIMIZATION
- ISYE 8813 - CONSTRAINT PROGRAMMING

## APPLICATION EL[illegible]ECTIVES

- CP 6514 - INTRODUCTION [illegible] SYSTEMS
- CP 6521 - ADVANCED GEO[illegible]
- CP 6541 - ENVIRONMENTA[illegible]
- CP 6570 - SOCIOECONOMIC GIS
- CP 6581 - PROGRAMMING FOR GIS

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- CP 8853 - CLIMATE CHANGE ANALYTICS
- CS 6440 - INTRO TO HEALTH INFORMATICS
- ISYE 6201 - MANUFACTURING SYSTEMS
- ISYE 6202 - WAREHOUSING SYSTEMS
- ISYE 6203 - TRANSPORTATION AND SUPPLY CHAIN SYSTEMS
- ISYE 6320 - PUBLIC IMPACT APPLICATIONS OF OPERATIONS RESEARCH AND MANAGEMENT SCIENCE
- ISYE 6335 - SUPPLY CHAIN ENGINEERING I
- ISYE 7201 - PRODUCTION AND SERVICE SYSTEMS ENGINEERING
- ISYE 7203 - LOGISTICS SYSTEMS ENGINEERING
- ISYE 8803 - MODEL IN MED DEC & HLTH POLICY
- ISYE 8813 - LIFE CYCLE AS



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Georgia Institute of Technology

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**H. Milton Stewart School of Industrial and Systems Engineering**
55 5th Street NW
Atlanta, GA 30332-0205
404.894.2300
Directions

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